

Solution Architecture

SMARTBRIDGE / SKILL WALL

Date: 29 September 2026

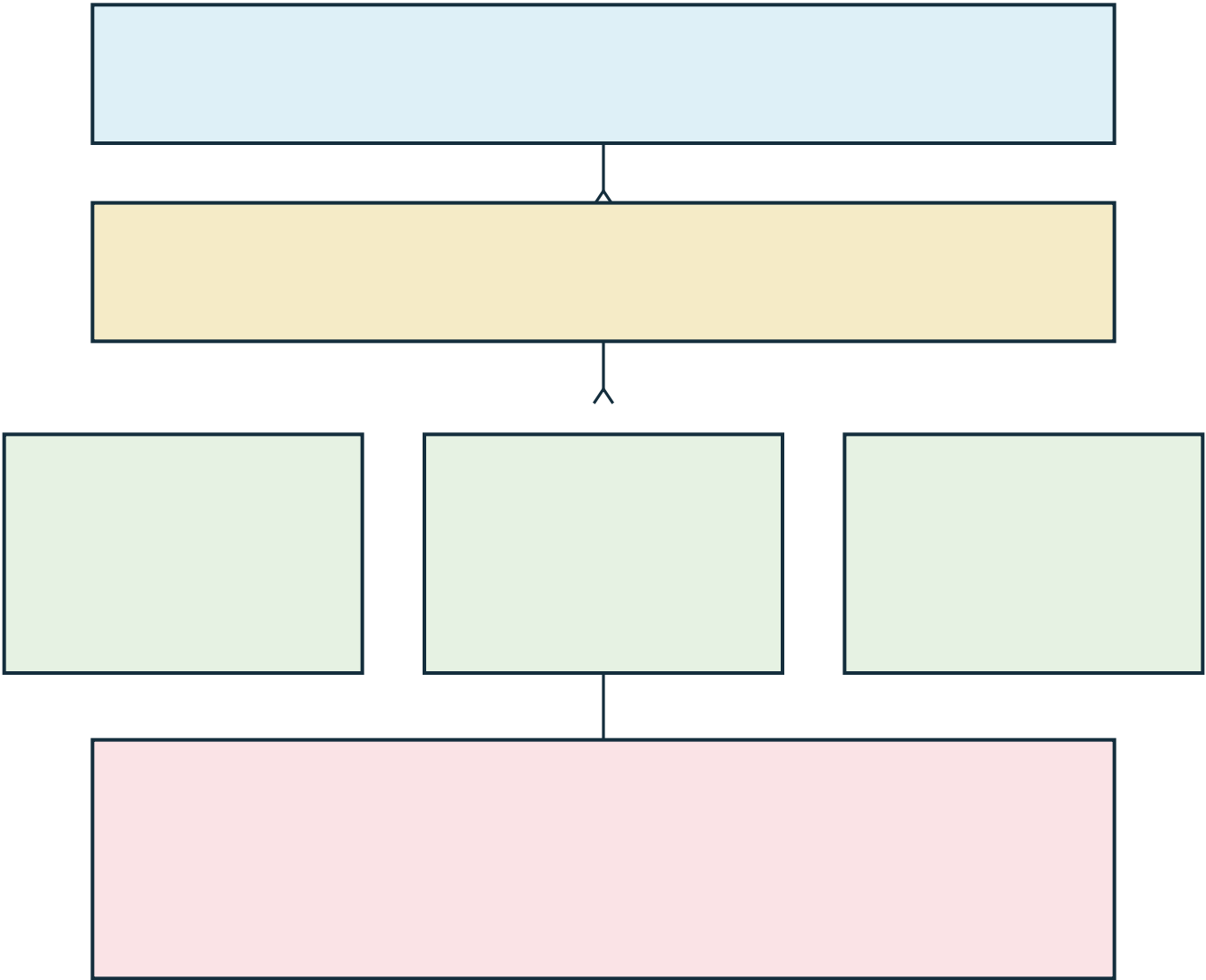
Team ID: SWTID-2026-3580

Project Name: EduGenie: Google Gemini Powered Learning Assistant

Maximum Marks: 5 Marks

Solution Architecture Diagram:

Fill the structural boxes with EduGenie's actual technologies and components.



Solution Architecture

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Component Description Table

Presentation Layer	Student-facing interface for learning tasks, inputs, and generated results.	HTML, CSS, JavaScript
API Gateway	Receives requests, validates input, routes tasks, returns responses.	FastAPI, Uvicorn
Auth Service	No user login/authentication in the MVP; Gemini credential remains server-side.	python-dotenv
Core Logic Service	Coordinates Q&A, explanation, quiz, summary, and learning-path modules.	Python, FastAPI
External API Integrations	Generative AI service used for the major learning workflows.	Google Gemini API
Local AI Model	Provides local concept explanation without requiring a remote call for the model.	llama3.1, LLaMini-Plan-T5-783M, Transformers, PyTorch
Data / Storage Layer	No persistent database in current MVP; project/runtime files and local model artifacts.	datasetbridge / model files
Architecture Notes		